A NUMERICAL INVESTIGATION OF THE INTER-LAYER BOND AND SURFACE ROUGHNESS DURING THE YIELD STRESS BUILDUP IN WET-ON-WET MATERIAL EXTRUSION ADDITIVE MANUFACTURING

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ABSTRACT

Performance of wet-on-wet material extrusion additive manufacturing (MEX-AM) requires attention to the printed layer before printing the next layer on top. This is for instance the case of thermoset resins that require curing, or concrete, which hardening reaction may take minutes to hours. In this work, a computational rheology model is used to simulate wet-on-wet MEX-AM. The model includes the partial curing/hardening of the printed layer by a yield stress buildup. The results highlight how the yield stress buildup of the previously deposited layers affects the inter-layer bond and surface roughness.

INTRODUCTION

Wet-on-wet printing of materials has recently gained attention when it comes to 3D printing different scales/sizes of components/structures. Materials such as thermoset, ceramic paste, and concrete can be printed wet-on-wet in large-scale using Robocasting and 3D concrete printing technologies [1-4]. These are freeform fabrication techniques under the umbrella term Material Extrusion Additive Manufacturing (MEX-AM). In these technologies, materials are extruded through a nozzle following a layer-by-layer approach similar to Fused Deposition Modelling [5-7].

Wet-on-wet printing can lead to unwanted deformation of previously deposited layers when printing on top of them [8]. Furthermore, larger deformation could negatively affect the surface roughness of layers. Therefore, it can be important to let a printed layer partially solidify before depositing the next layer on top. This is simply equivalent to allowing a yield stress buildup of the printed layer [9]. A higher yield stress buildup could potentially solidify the printed layer that negatively affects the interlayer bond [2]. However, it requires an investigation on how much yield stress buildup is necessary to achieve a homogeneous layer size as well as have a better inter-layer bond and surface roughness.

In this work, we developed a computational rheology model that simulates a two-layer structure by MEX-AM. Both layers are modelled with different yield stress to represent the printing of a wet layer onto a partially solidified bottom layer. The computational model provides the cross-sections of the deposited layers, from which the bottom layer deformation, the inter-layer bond, and the surface roughness were estimated.

COMPUTATIONAL MODEL

The computational fluid dynamics (CFD) model developed in ***FLOW-3D***® (Version 12.0; 2019; Flow Science, Inc.) [10] solves the transient and isothermal deposition flow of materials using the continuity and momentum balance equations given below:

$$\nabla . \boldsymbol{u} = 0 \quad (1)$$

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u}.\nabla\boldsymbol{u}\right) = \rho\boldsymbol{g} - \nabla p + \nabla.\boldsymbol{S}_T \quad (2)$$

where $\boldsymbol{u}$ is the velocity vector, ρ =1000 kg.m^{-3} is the constant density, p is the pressure, $\boldsymbol{g}$ is the gravitational acceleration vector with components $(0,0,-g)$, t is the time, and $\boldsymbol{S}_T$ the deviatoric stress-tensor. The $\boldsymbol{S}_T$ is modelled within the following generalized Newtonian fluid constitutive framework:

$$\boldsymbol{S}_T = 2\mu_{APP}\,(\dot{\gamma})\,\boldsymbol{D}_T \quad (3)$$

Where $\dot{\gamma} = \sqrt{2\mathrm{tr}(\boldsymbol{D}_T^2)}$ is the strain rate magnitude, $\boldsymbol{D}_T = \left((\nabla\boldsymbol{u}) + (\nabla\boldsymbol{u})^T\right)/2$ is the deformation rate tensor, and μ_{APP} is the apparent viscosity of the viscoplastic material. The viscoplastic nature is modelled by the regularized bi-viscous Bingham model [11]. In addition, the model uses a scalar approach to differentiate the material of layers and assign different yield stress. Details of the scalar approach can be found in [9], whereas a simple representation together with the bi-viscous Bingham model is given in Equation 4 that identifies the printed material by the presence of a non-dimensional scalar identifier S. The two layers have different yield stresses, resulting in a different apparent viscosity. The printing layer uses Equation 4 with $S = 0$ and it is the printed bottom layer when $S = 1$.

$$\mu_{APP} = (1-S)\mu_{ap}(\dot{\gamma}) + S\mu_{printed.ap}(\dot{\gamma}) = \begin{cases} \eta_p + \dfrac{(1-S)\tau_{printing} + S\tau_{printed}}{\dot{\gamma}_c}, & \text{for } \dot{\gamma} < \dot{\gamma}_c \\ \eta_p + \dfrac{(1-S)\tau_{printing} + S\tau_{printed}}{\dot{\gamma}}, & \text{for } \dot{\gamma} \geq \dot{\gamma}_c \end{cases} \quad (4)$$

Where μ_{ap} and $\mu_{printed.ap}$ are the apparent viscosity of the printing and printed layers, respectively, $\eta_P = 33$ Pa.s is the plastic viscosity, $\tau_{printing} =$ 100 Pa is the printing yield stress, $\tau_{printed}$ is the printed yield stress, which was varied between 100 Pa to 400 Pa, and $\dot{\gamma}_c$ =0.05 s^{-1} is the critical shear rate.

The computational domain of the CFD model includes a cylindrical nozzle of diameter 25 mm that extrudes the material with an extrusion speed of 40 mm/s, cf. Figure 1. The nozzle moves with a printing speed of 50 mm/s. The nominal layer height is 12.5 mm. A uniform Cartesian grid is used to mesh the computational domain that is discretized by the Finite Volume Method (FVM). The top and bottom plane of the domain is an inlet and wall boundary, respectively. A symmetrical boundary is assigned through the nozzle diameter along the printing direction. Other boundaries are assigned to a continuative boundary condition. Moreover, the open channel flow is modelled with the Volume of Fluid Method (VOF) [12]. The pressure and velocity components are solved implicitly in time. The viscous solver utilizes the scalar identifier based on a self-customization to alter the viscosity between the layers. Figure 2 illustrates the dynamic viscosity distribution. It can be seen that the model predicts the yield stress buildup at the bottom layer accurately (bottom).

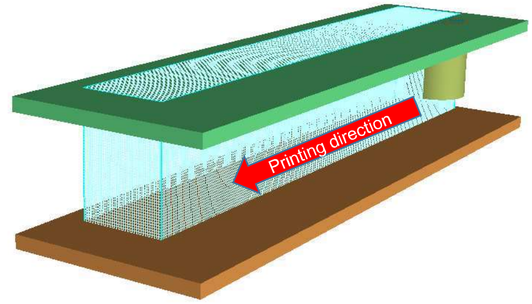

FIGURE 1. CFD model with the computational domain, extrusion nozzle, and substrate.

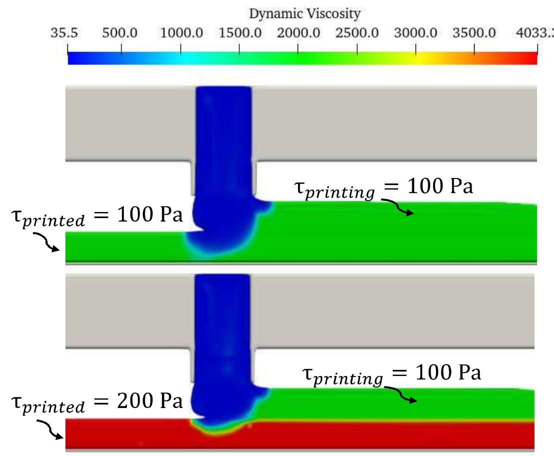

FIGURE 2. Dynamic viscosity distributions between the layers for pure wet-on-wet printing (top) and yield stress buildup during wet-on-wet (bottom).

The simulated results are post-processed in order to obtain the cross-sectional shapes. The cross-sections are taken in the middle of the computational domain.

RESULTS AND DISCUSSION

The cross-sectional shapes of the deposited layers and inter-layer bond lines are presented in Figure 3. It can be seen that the shape of the bottom layer deviates less when the yield stress increases. Furthermore, the bond lines can be observed to move from concave to convex state with the buildup. It is also seen that the bond lines move their position from below the initial height of the bottom layer to the exact position of the initial height. These behaviors are seen because the yield stress increased before the load of the second layer was applied. As a result, both the deformation of the bottom layer (i.e., ratio between the bottom layer width after printing the second layer and width at the end of the first layer, W1) and inter-layer bond (i.e., ratio of the length of bond line and W1) decreases with yield stress buildup, cf., Figure 4. Although it is a mild decrease for inter-layer bonds, and for higher yield stress buildups (i.e., $\tau_{printed}$ = 300 and 400 Pa), it is almost identical.

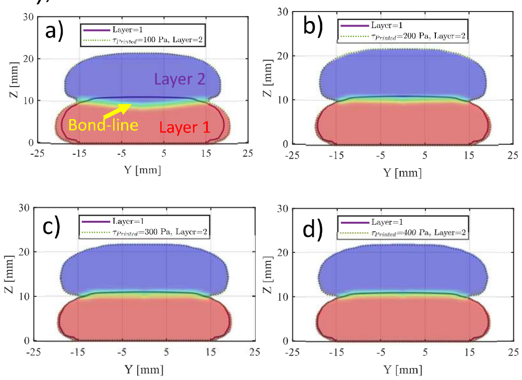

FIGURE 3. Cross-sectional shapes and inter-layer bond lines for different yield stress buildup. Moreover, a) $\tau_{printed}$ =100 Pa; b) $\tau_{printed}$ =200 Pa; c) $\tau_{printed}$ =300 Pa; d) $\tau_{printed}$ =400 Pa.

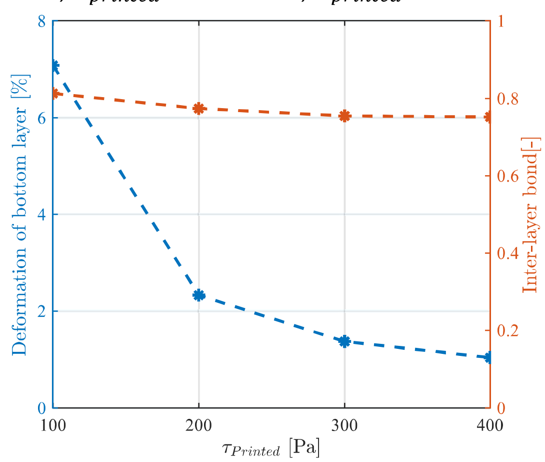

FIGURE 4. Deformation of the bottom layer and inter-layer bond for different yield stress buildup.

Figure 5 illustrates the surface roughness (i.e., edge smoothness and arithmetic mean roughness Ra) of the vertical edge profile of the simulated part for different $\tau_{printed}$. The profile smoothness is represented by the centerline positioned where the areas above and below the line are equal, and the arithmetic mean roughness is estimated as the mean deviation of the profile from the centerline, cf. [13] for details. It is seen that the yield stress buildup improves the surface quality as well as makes the vertical edges smoother. However, Ra is almost the same for larger $\tau_{printed}$ = 300 and 400 Pa.

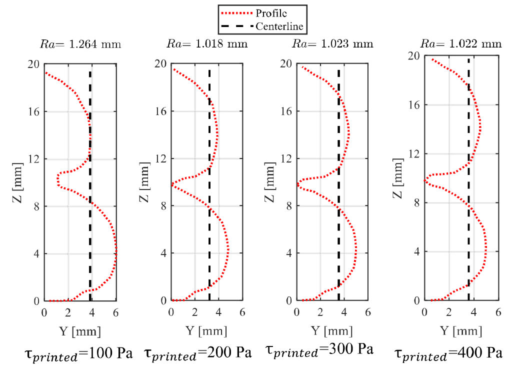

FIGURE 5. Surface roughness for different yield stress buildup, $\tau_{printed}$.

SUMMARY

A computational fluid dynamics model is used to predict the deposition flow of viscoplastic materials during the yield stress buildup in wet-on-wet MEX-AM. The cross-sectional shapes of the layers are predicted to investigate the bottom layer's deformation, interlayer bond, and surface roughness of layers.

The yield stress buildup at the bottom layer reduces its deformation and improves the surface roughness and smoothness. On the other hand, it can be detrimental for the inter-layer bonding when the buildup is too high as it slightly decreases with buildup. Thus, an optimal balance in the yield stress buildup is required as depicted in this study.

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